TÉCNICO LISBOA

Computational Fluid Mechanics

Masters in Aerospace Engineering

Sod's shock tube & M5-SWBLI

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Contents

1 Sod's shock tube **1**
- 1.1 Introduction . . . 1
- 1.2 Theoretical Background . . . 1
  - 1.2.1 Governing Equations . . . 1
  - 1.2.2 Characteristics . . . 1
  - 1.2.3 Analytical Solution . . . 1
- 1.3 Numerical Setup . . . 2
  - 1.3.1 Domain and Mesh . . . 2
  - 1.3.2 Physical Models . . . 2
- 1.4 Results and Discussion . . . 2
  - 1.4.1 Comparison with Analytical Solution . . . 2
  - 1.4.2 Influence of the Riemann Solver . . . 3
- 1.5 Conclusion . . . 3

2 Mach 5 shock wave/boundary layer interaction: M5-SWBLI **4**
- 2.1 Introduction . . . 4
- 2.2 Theoretical Background . . . 4
  - 2.2.1 Shock Wave . . . 4
  - 2.2.2 Boundary Layer . . . 5
  - 2.2.3 SWBLI . . . 5
- 2.3 Numerical Setup . . . 5
  - 2.3.1 Domain and Mesh . . . 5
  - 2.3.2 Physical Models . . . 6
- 2.4 Results and Validation . . . 6
- 2.5 Conclusion . . . 8

Appendices **10**

1 Sod's shock tube

1.1 Introduction

This report presents the numerical solution and verification of the classic Sod's Shock Tube problem, a standard benchmark for compressible flow solvers introduced by Gary Sod in 1978. The objective is to solve the Euler equations for an ideal gas in a 1D domain and compare the numerical results obtained using STAR-CCM+ against the exact analytical solution at time $t = 0.01$ s.

The problem consists of a tube divided by a diaphragm at $x = 0$. At $t = 0$, the diaphragm bursts, separating a region of high pressure (Region 1) from a region of low pressure (Region 5).

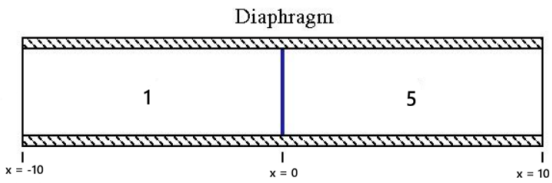


Figure 1: Sketch of the tube with the Diaphragm at t = 0s

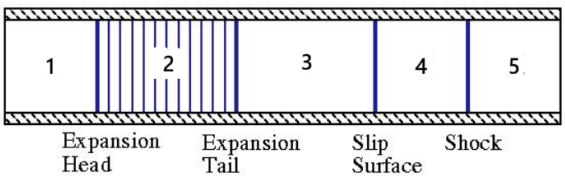


Figure 2: Sketch of the tube without the Diaphragm at t = 0.01s

The pressure ratio is $p_1/p_5 = 10$. The computational domain extends from $x = -10$ m to $x = 10$ m.

1.2 Theoretical Background

1.2.1 Governing Equations

The flow is governed by the 1D Euler equations for an ideal gas. As discussed in the course lectures (Lecture 24), the conservation laws can be expressed in the differential form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} = 0 \tag{1}$$

However, strictly speaking, the differential form is not valid across discontinuities. Therefore, for problems involving shock waves like the Sod tube, we must rely on the **Integral Form** over a control volume $\Omega \equiv [x_L, x_R]$:

$$\frac{d}{dt}\int_{\Omega} \mathbf{U}\, dV = -[\mathbf{F}(\mathbf{U}_R) - \mathbf{F}(\mathbf{U}_L)] \tag{2}$$

This formulation allows for "weak solutions" (shocks and contact discontinuities) where the derivatives are undefined.

1.2.2 Characteristics

The hyperbolic nature of the system is defined by its eigenvalues or characteristic speeds λ:

$$\lambda_1 = u - a, \quad \lambda_2 = u, \quad \lambda_3 = u + a \tag{3}$$

where a is the speed of sound. As stated in the lecture notes, "information cannot travel faster than the speed of sound".

- **Shock Wave:** Formed when characteristics intersect ($|u| > a$ locally), leading to "information destruction" and entropy generation.
- **Rarefaction Wave:** A region where characteristics diverge, acting as a "source of fluid" while maintaining constant entropy.

1.2.3 Analytical Solution

For the analytical solution the fortran file provided in the study case page was adapted into a matlab file for easier handling of the data. The following initial conditions were considered.

Table 1: Analytical Solution ($t = 0$ s)

Prop	Region 1	Region 5
p [Pa]	$1 \cdot 10^5$	$1 \cdot 10^4$
u [m/s]	0	0
ρ [kg/m^3]	1	0.125
T [K]	348.43	278.74

1.3 Numerical Setup

The simulation was performed using STAR-CCM+. The 1D behavior was replicated using a quasi-1D grid.

1.3.1 Domain and Mesh

- **Domain:** $x \in [-10, 10]$ m.
- **Discretization:** The axial direction was discretized with $N = 200$ cells ($\Delta x = 0.4$ m).
- **Boundaries:** *Symmetry Plane* (or Slip Wall) on lateral surfaces to ensure 1D inviscid flow.

Instead of 50 cells in the axial direction of the tube it was chosen to use 200 as it would not only improve the quality of the results but also because it would be easier to work with the mesh that way as the cells were perfectly divided in the tube and its centroids were at equal distances.

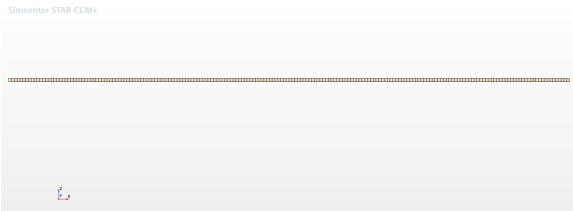


Figure 3: Computational Mesh

1.3.2 Physical Models

- **Time:** Implicit Unsteady.
- **Number of inner iterations:** 15
- **Maximum Physical time:** 0.01s
- **Material:** Ideal Gas.
- **Flow:** Coupled Flow.
- **Viscous Regime:** Inviscid (Euler).

Initial conditions were initialized using *Field Functions* to define the step change at $x = 0$.

1.4 Results and Discussion

1.4.1 Comparison with Analytical Solution

The numerical results at $t = 0.01$ s obtained with STAR-CCM+ are compared with the analytical solution below.

Analysis of Flow Features: With the coarse mesh ($N = 200$), the results exhibit typical behaviors of numerical approximations:

1. **Shock Smearing:** The shock wave at $x \approx 5.54$ m is not a perfect discontinuity but is smeared over 2-3 cells. This is due to the *Artificial Dissipation* (implicit in the upwind scheme) which mimics physical viscosity to prevent numerical instability (Lecture 23/24).

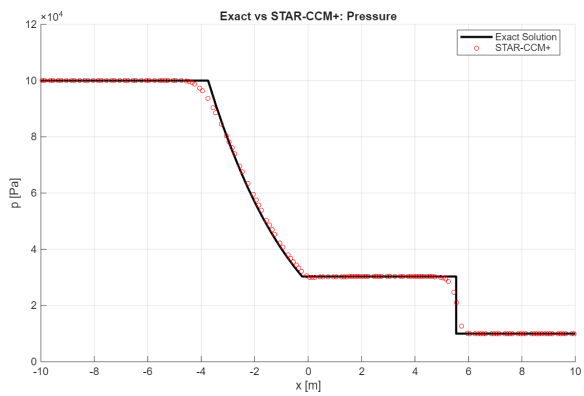


Figure 4: Comparison between the Pressure distribution in the Analytical and the Numerical solution.

2. **Contact Discontinuity:** The density jump at $x \approx 2.93$ m shows more diffusion than the shock. Unlike shocks, contact discontinuities do not have a self-sharpening mechanism (characteristics run parallel to the interface, $\lambda = u$), making them highly sensitive to numerical diffusion.

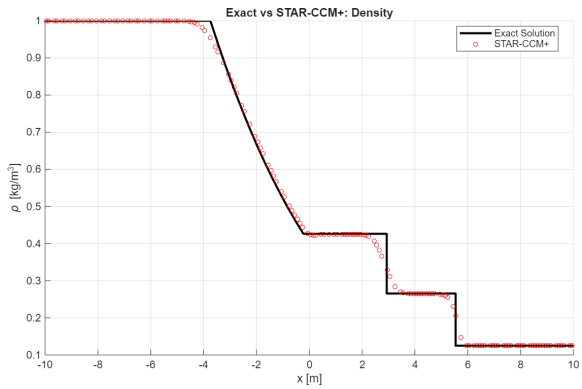


Figure 5: Comparison between the Density distribution in the Analytical and the Numerical solution.

1.4.2 Influence of the Riemann Solver

The simulation uses the STAR-CCM+ "Coupled Flow" model, which relies on an Approximate Riemann Solver (e.g., Roe or AUSM+).

- **Godunov vs. Approximate Solvers:** As noted in the lectures, employing an exact Godunov solver is computationally expensive. Approximate solvers linearize the Riemann problem at each cell face.

- **Entropy Condition:** A critical aspect is satisfying the "Entropy Condition" (Lecture 24) to ensure only physically valid shocks are captured. The results confirm that the solver correctly captures the rarefaction wave (Region 2) without unphysical expansion shocks.

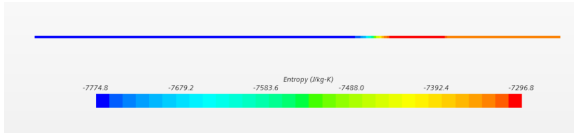


Figure 6: Entropy distribution along the tube.

1.5 Conclusion

The shock tube problem was successfully simulated. In the comparison the 5 regions are clearly visible as expected in both the numerical anthe analytical solutions. The comparison verifies that the code correctly solves the Euler equations. The deviations observed are consistent with the discretization error introduced by the coarse mesh ($N = 200$), confirming the importance of grid refinement and appropriate flux schemes for capturing high-gradient flow features.

2 Mach 5 shock wave/boundary layer interaction: M5-SWBLI

2.1 Introduction

This work addresses the NASA Mach 5 shock wave/boundary-layer interaction (M5-SWBLI) validation case by solving the governing equations in STAR-CCM+ for a shock-generator/flat-plate configuration.

SWBLI is a canonical compressible-flow problem in supersonic and hypersonic regimes. It is directly relevant to the design of high-speed aerodynamic components such as inlets and control surfaces. Shock-induced adverse pressure gradients can strongly modify turbulent boundary layers and can trigger flow separation.

In high-speed inlet environments, SWBLI is often categorized as (quasi) two-dimensional when generated by configurations such as compression corners or impinging/reflected shock waves, and three-dimensional when induced by elements such as sharp or blunt fins, swept compression corners, or double-fin arrangements that create crossing-shock patterns. Impinging/reflected shocks are among the easiest cases to study this interaction with, with the main driving parameter being the shock strength rather than the geometry of the shock generator [3].

In the present study, the interaction is investigated using the geometry shown in Fig. 7, where the shock generator is set to $\beta = 14°$. Here, β is the angle of the shock generator with respect to the horizontal (incoming freestream direction). This angle sets the oblique-shock strength and location, and therefore the interaction with the turbulent boundary layer developing along the flat plate.

Figure 7: Shock generator and flat plate geometry

In particular, the objective of this test is to extract, from the converged CFD solution, key wall and flow quantities such as the Mach-number distribution and the skin-friction coefficient C_f. Skin friction (and the corresponding coefficient C_f) directly quantifies wall shear and boundary-layer behaviour, and it is especially sensitive to shock-induced separation and reattachment [2]. For this reason, skin-friction measurements are widely used to validate turbulence closures and near-wall modelling in CFD, especially in the presence of strong adverse pressure gradients and compressibility effects. In this work, the Spalart–Allmaras (SA) and SST (k–ω SST) turbulence models are compared based on the predicted Mach-number field and the streamwise development of C_f.

2.2 Theoretical Background

2.2.1 Shock Wave

Mach number M of magnitude > 1 (*supersonic* flow), determined by the relation

$$M = \frac{U}{a}, \tag{4}$$

where U is the flow velocity and a is the speed of sounds, can lead to the formation of shock waves. These are thin regions characterized by abrupt changes in flow properties - such as the Mach number. At Mach 5, the flow is considered *hypersonic*.

2.2.2 Boundary Layer

The no-slip condition of a viscous wall generates a *boundary layer*, where velocity transitions from zero at the wall to the freestream velocity. The presence of such boundary layer leads to strong velocity gradients near the wall. The wall shear stress τ_w at the boundary plays a fundamental role in determining the skin-friction drag. The skin friction coefficient C_f can be calculated via

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2}, \tag{5}$$

where ρ_∞ is the freestream density and U_∞ is the freestream density. In turbulent boundary layers, higher skin friction is seen compared to laminar flows [1].

2.2.3 SWBLI

Shock wave/boundary layer interaction (or SWBLI) leads to a sudden rise in pressure and a strong adverse pressure gradient. As a result, flow separation occurs, leading to a region of inverse flow (an area of *recirculation*).

2.3 Numerical Setup

2.3.1 Domain and Mesh

The computational domain is built around the geometry shown in Fig. 7. It consists of a shock-generator plate, 300 mm long, set at a wedge angle $\beta = 14°$. The resulting oblique shock impinges on a flat plate of length 500 mm.

The final mesh used for the simulations is shown in Fig. 8. A two-dimensional unstructured grid is adopted, and prism layers are applied along the viscous wall to resolve the near-wall region. Local refinement is introduced in the regions where the strongest gradients are expected, in particular around the shock-generator leading edge and in the shock-boundary-layer interaction zone.

Figure 9 highlights the refinement near the junction between the inviscid upper wall and the viscous shock-generator surface. This junction coincides with the change in wall orientation that generates the oblique shock. The grid is also refined downstream in the interaction region, where boundary-layer separation and flow recirculation may occur. The final mesh contains a total of 32,391 cells.

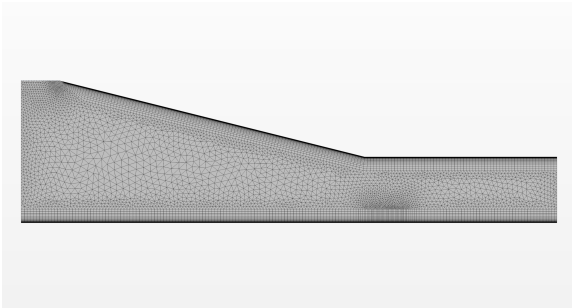


Figure 8: Computational Mesh

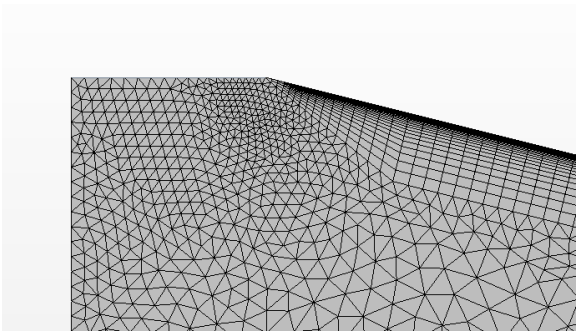


Figure 9: Close-up of the mesh refinement.

The boundary conditions applied to the computational domain are shown in Fig. 10. A uniform freestream condition is prescribed at both the inlet (1) and the outlet (2). In the present setup, the inlet freestream is defined using static quantities, namely $p_\infty = 4000$ Pa (with the reference pressure set to 0 Pa), $T_\infty = 70$ K, and a streamwise velocity $U_\infty = 838$ m/s. These values correspond to a freestream Mach number close to $M_\infty \approx 5$. The outlet is treated as a freestream boundary to allow disturbances to leave the domain with minimal reflection.

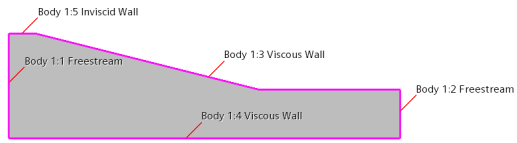


Figure 10: Boundary conditions assignment

The upper-left boundary (5) is modelled as an inviscid (slip) wall, enforcing no penetration and zero tangential shear stress. This choice avoids the formation of an additional boundary layer on the upper boundary. The shock-generator surface (3) and the flat plate (4) are modelled as viscous (no-slip) walls. Prism layers are applied along the viscous walls to resolve near-wall gradients and to capture the shock–boundary-layer interaction on the flat plate. Wall thermal conditions are imposed according to the assignment specifications (isothermal walls at $T_w = 300$ K). The inflow turbulence quantities are set consistently with the selected turbulence model (SA or SST).

All simulations are performed as steady, two-dimensional, compressible RANS computations in STAR-CCM+. The governing equations are solved using a coupled flow and coupled energy formulation, with air modelled as an ideal gas.

Turbulence is modelled using eddy-viscosity RANS closures. Two models are considered: the one-equation Spalart–Allmaras (SA) model and the two-equation k–ω SST model.The same domain, mesh, boundary conditions, and numerical settings are used for both turbulence models, so that differences in the results can be attributed to the turbulence closure.

2.3.2 Physical Models

For the simulation, the SST and SA turbulence models were used in the following order: Two Dimensional, Steady, Coupled Flow, Gradients, Ideal Gas, Coupled Energy, Turbulent, Reynolds-Averaged Navier-Stokes, Solution Interpolation. Then,

- *for the SST simulation*: K-Omega Turbulence, SST (Menter) K-Omega, Wall Distance, All y+ Wall Treatment.
- *for the SA simulation*: Spalart-Allmaras Turbulence, High-Reynolds Number Spalart-Allmaras, Wall Distance, High y+ Wall Treatment.

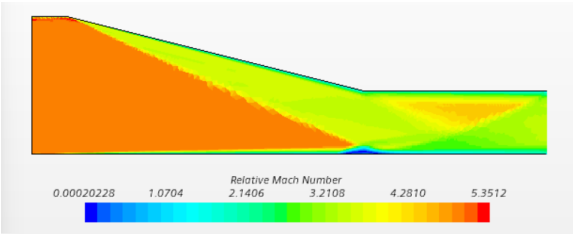
To plot the C_f on the bottom boundary (viscous wall), a new field function was defined (the pre-defined STAR CCM+ *Skin Friction Coefficient* field function takes in absolute values of shear stress, which would not allow to show the recirculation area properly). The following was defined:

- *Skin Friction Coefficient Signed*
  `dot($$WallShearStress,[1,0,0])/ (0.5 * $densityref * pow($Uref_3, 2))`
- *densityref*
  `0.204`
- *Uref_3*
  `828`

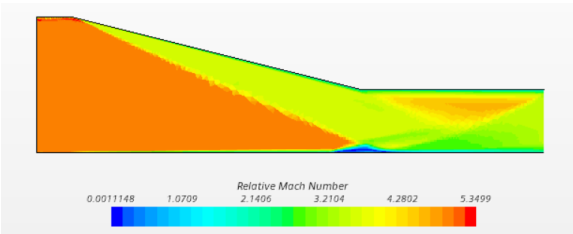
2.4 Results and Validation

The numerical results are validated by comparison with the experimental database reported in the DLR technical report by Schülein *et al.* [3] and with the optical skin-friction measurement methodology and reference data discussed by Schülein [2].

The results (Mach number contours) obtained with the SA and SST turbulence models are shown in Fig. 11, while the corresponding residual histories are reported in Fig. 12.

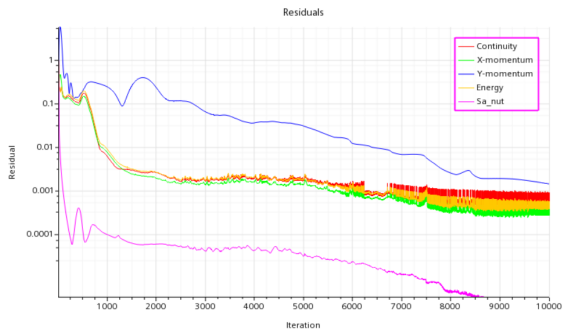


(a) SA turbulence model.

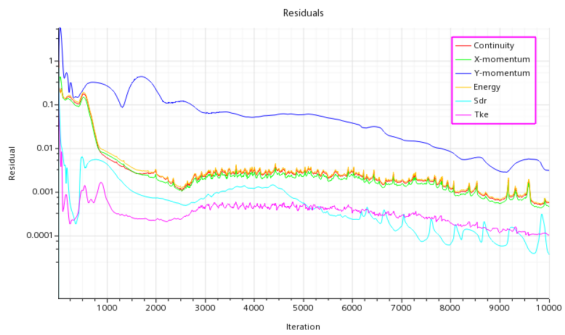


(b) SST turbulence model.

Figure 11: Comparison of Mach-number contours for different turbulence models.

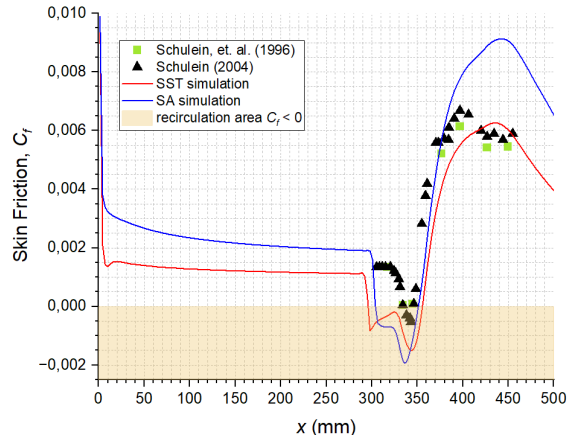


(a) SA turbulence model.

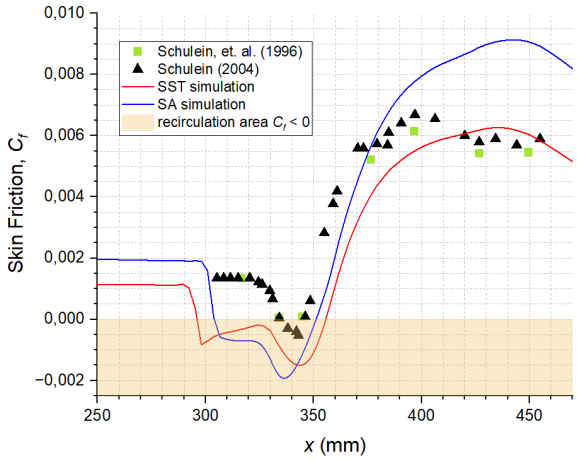


(b) SST turbulence model.

Figure 12: Comparison of the residuals for different turbulence models.



(a) The whole domain.



(b) Close up on the recirculation area.

Figure 13: The skin friction coefficient on the bottom viscous wall. Green and black - experimental data. Blue and red - simulation. Yellow - recirculation area.

In both simulations, the residuals decrease with iteration but do not reach very low levels, even after a large number of iterations. In the present case, the convergence difficulty appears to be associated with the flow behaviour close to the upper inviscid (slip) wall. In particular, the Mach-number field, which is expected to remain approximately uniform in the freestream upstream of the interaction, shows a non-uniform distribution near the slip wall, with local gradients and localized peaks.

A plausible explanation is the combina-

tion of (i) strong compressible-wave structures in the vicinity of the upper boundary and (ii) insufficient mesh resolution and/or abrupt size transitions near the junction between the inviscid wall and the viscous shock-generator surface. A possible improvement would be to refine the mesh locally around this corner, more than it is already refined, and along the expected shock path.

Despite the above issues, the main flow features are captured. The Mach-number contours show the formation of the oblique shock and its interaction with the boundary layer on the lower wall, and the wall skin-friction coefficient C_f exhibits the expected response in the interaction region.

In Fig. 13 the behaviour of the skin friction coefficient on the bottom viscous wall is presented. The SA model shows more extreme values than the SA model (higher values for $C_f > 0$ and lower values for $C_f < 0$). Both models show earlier departure into the recirculation area than the experimental results. The SA solution predicts a smaller separation bubble than the SST model, which is consistent with the assumptions. The area where the skin friction coefficient decreases below zero corresponds to the region of the shock wave/boundary layer interaction.

2.5 Conclusion

The M5-SWBLI was simulated using the STAR-CCM+ software. Mach number contours and the skin-friction coefficient at the bottom wall was observed. The SST and SA turbulence models were used and compared. The turbulence models behave as expected, showing smaller recirculation area for the SA model. The negative C_f numbers are observed at the region where the shock wave interacts with the boundary layer, which is consistent with theory.

References

[1] Herrmann Schlichting and Klaus Gersten. *Boundary-Layer Theory*. Springer Berlin Heidelberg, 2000.

[2] Erich Schülein. Optical skin friction measurements in the short-duration ludwieg tube facility. In *Proceedings of the 20th International Congress on Instrumentation in Aerospace Simulation Facilities (ICIASF 2003 Record)*, pages 157–168, Göttingen, Germany, August 2003. IEEE.

[3] Erich Schülein, P. Krogmann, and E. Stanewsky. Documentation of two-dimensional impinging shock/turbulent boundary layer interaction flow. DLR-Interner Bericht DLR-IB 223-96 A 49, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Göttingen, Germany, October 1996.

Appendices

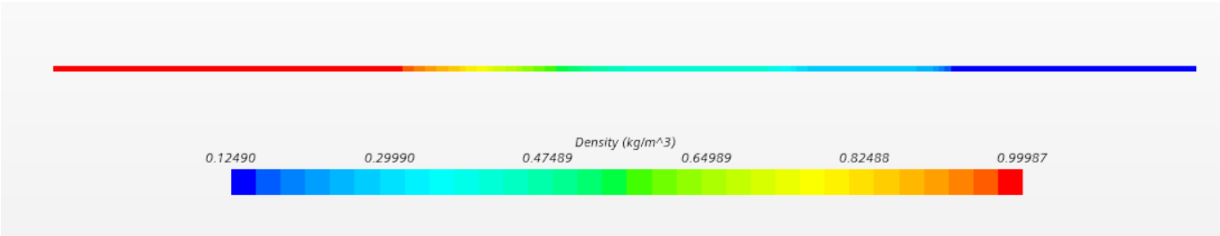


Figure 14: Density Distribution along the tube.

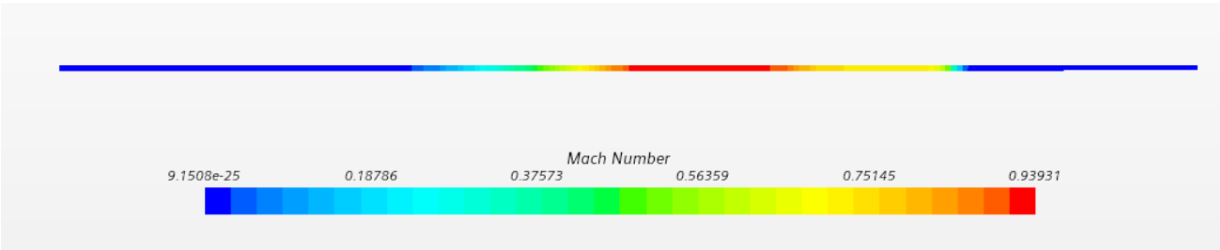


Figure 15: Mach Number Distribution along the tube.

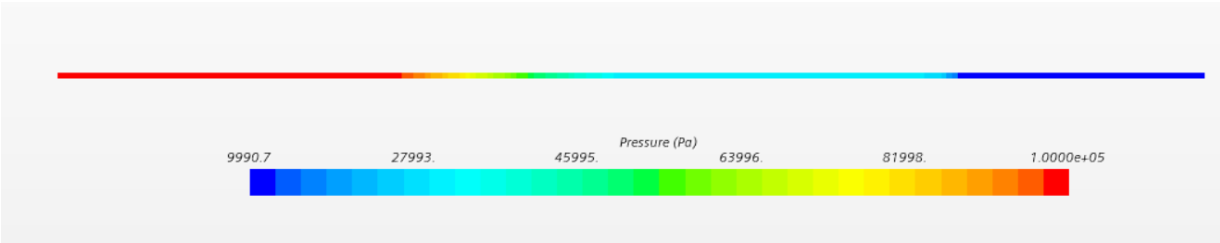


Figure 16: Pressure Distribution along the tube.

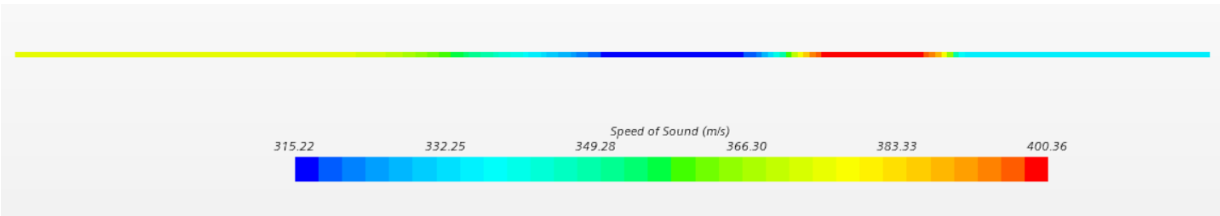


Figure 17: Speed of Sound Distribution along the tube.

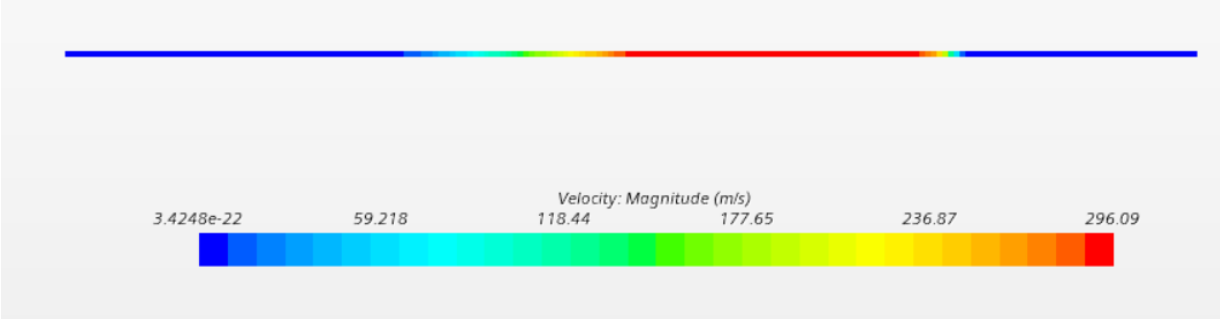


Figure 18: Velocity Distribution along the tube.

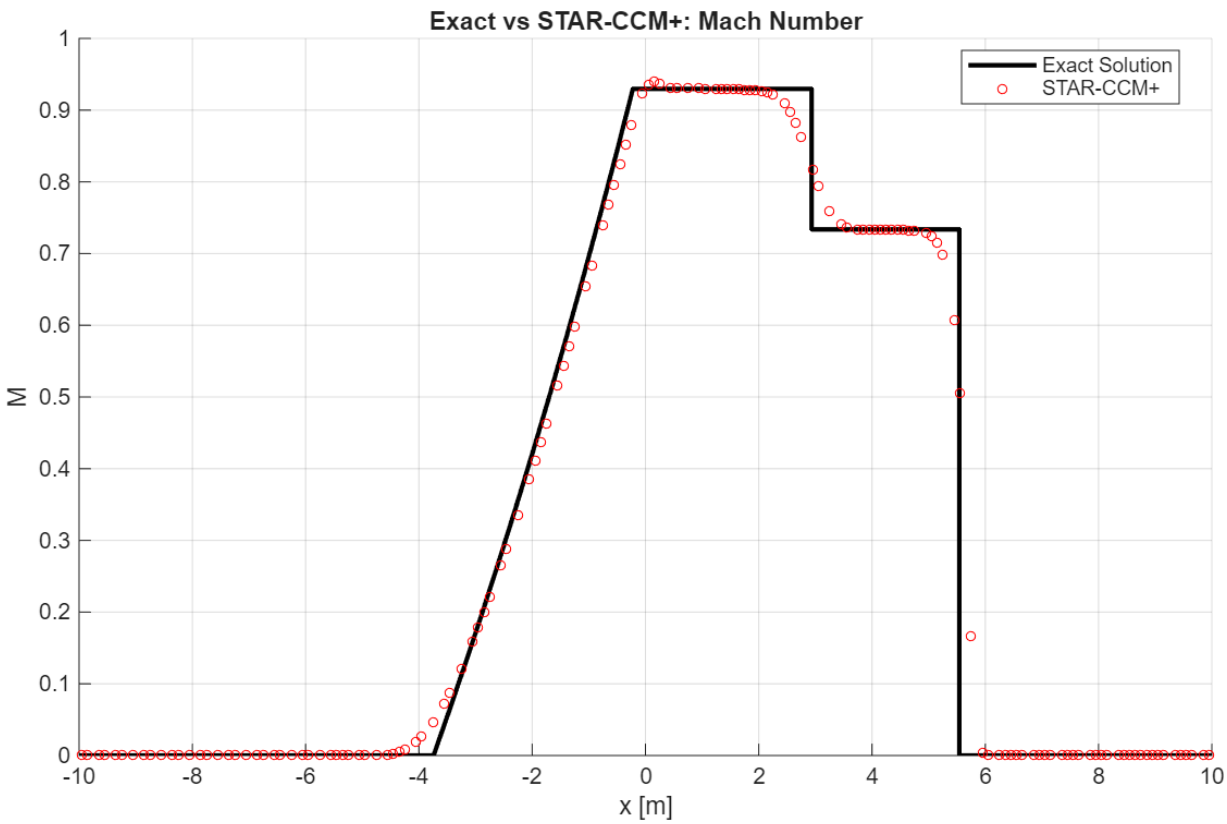


Figure 19: Comparison between the Mach Number distribution in the Analytical and the Numerical solution.

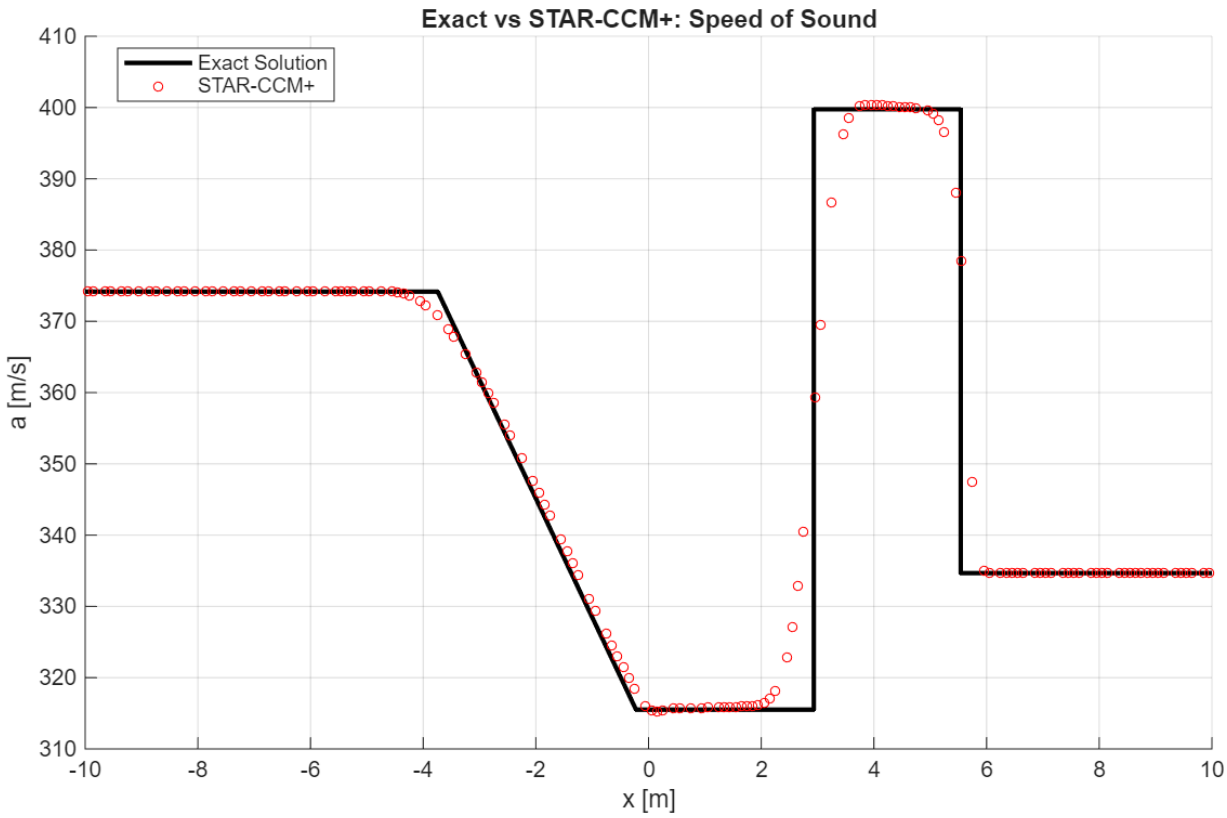


Figure 20: Comparison between the Speed of Sound distribution in the Analytical and the Numerical solution.

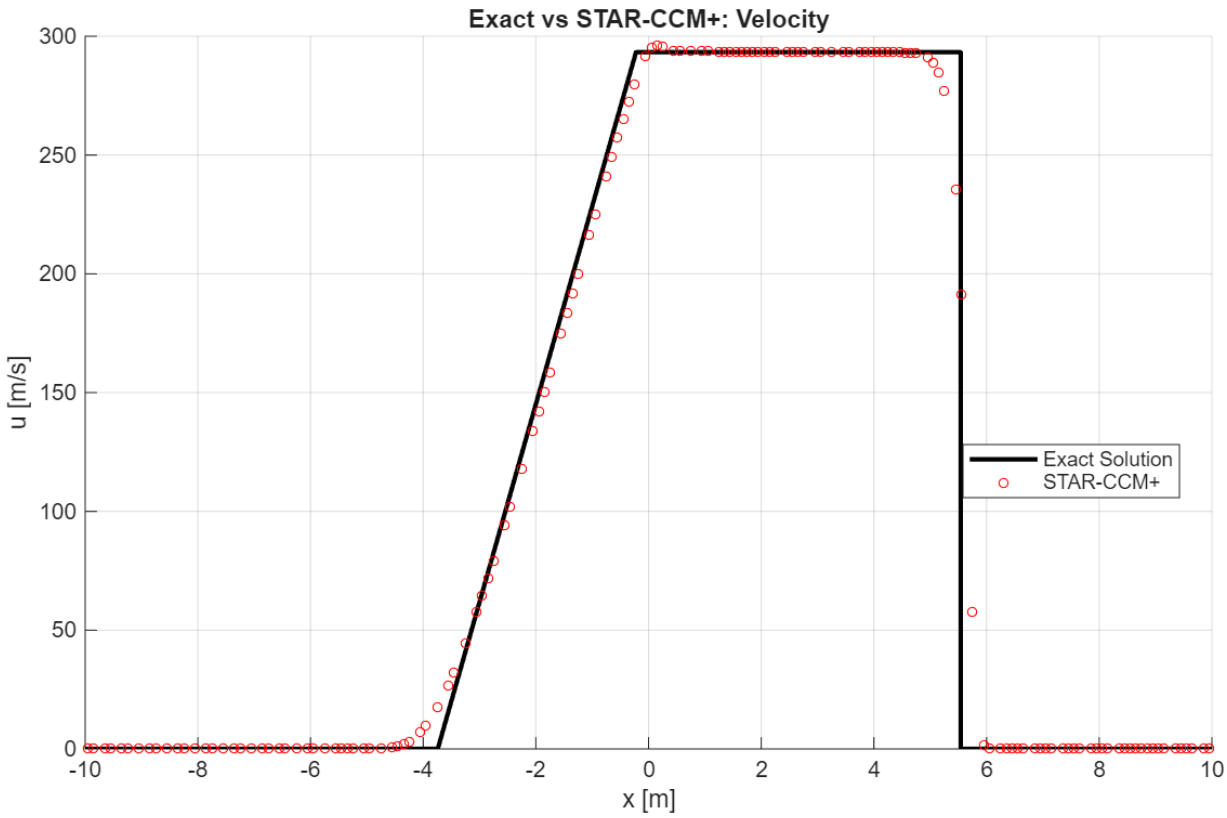


Figure 21: Comparison between the Velocity distribution in the Analytical and the Numerical solution.